CENG 252, Fall 2026, HW#3

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Due: 8:59pm, Oct 31, 2026

(Please turn in your work to Gradescope)

Problem 1 Read Fogler on the topic of PSSH and enzymatic reactions. (No turn-in requirement)

Problem 2

The following reaction proceeds with a nonelementary kinetics involving H• and Br• intermediates.

$$H_2 + Br_2 \rightarrow 2HBr$$

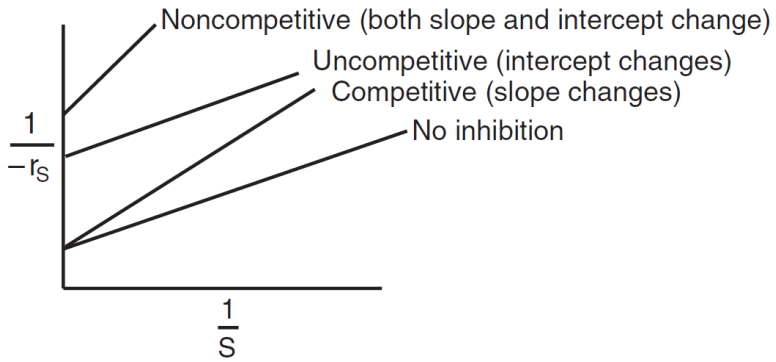
It is found that the reaction follows the below rate law:

$$r_{HBr} = \frac{k' C_{H_2} C_{Br_2}^{3/2}}{C_{Br_2} + k'' C_{HBr}}$$

Follow lecture slides note to propose the elementary steps and derive to the rate law step by step to match the proposed rate law. **Show your work step-by-step.**

Problem 3.

The Lineweaver–Burk plots for three types of enzyme inhibition looks like below. In the class, we have talked about competitive inhibition by deriving the rate law (see the lecture slides, we have a few pages to finish but conceptually it is similar). We also compared the slops and intercepts between Michaelis-Menten Kinetics and competitive inhibition. Please follow the same procedure, practice yourself and derive the rate law for Uncompetitive Inhibition and Non-competitive Inhibition. You may follow Fogler but you need to come to the modified Michaelis-Menten Kinetics in the reciprocal format so explain the L-B plots below. **You only need to show your work on Noncompetitive inhibition**.



Problem 4

The gas-phase epoxidation of butadiene over Ag catalysts has been measured to be 1st order in oxygen, 0th order in butadiene, and -1st order in epoxybutene.

$$C_4H_6 + \tfrac{1}{2}\, O_2 \rightarrow C_4H_6O$$

a. Devise a mechanism and rate determining step consistent with the observed rate law.

b. Assume that the reaction has a rate constant of 3.1×10^{-4} mol/s-bar-kg catalyst, and that the dominant term in the denominator of your Langmuir-Hinshelwood expression has a K value of 4.0 bar^{-1}. What will be the required catalyst mass in a CSTR-type reactor (i.e., a fluidized bed) to achieve a conversion of 30%? Assume an equimolar feed of O_2 and butadiene in a stream containing 75% cyclohexane at a temperature of 175°C and 30 bar, with an overall volumetric flow rate of 100 L/s. Assume ideal gas behavior.

Problem 5. Solving reaction mechanism below.

t-Butyl Alchohol (TBA) is an important octane enhancer that is used to replacee lead additives in gasoline. TBA was produced by the liquid-phase hydration (W) of isobutene (I) over an Amberlyst-15 catalyst. The system is normally a multiphase mixture of hydrocarbon, water and solid catalyst. However, the use of cosolvents or excess TBA can achieve reasonable miscibility. The reaction mechanism is beliveved to be the following:

$$I + S \Leftrightarrow I \bullet S$$

$$W + S \Leftrightarrow W \bullet S$$

$$W \bullet S + I \bullet S \Leftrightarrow TBA \bullet S + S$$

$$TBA \bullet S \Leftrightarrow TBA + S$$

Derive a rate law assuming:

(a) The surface reaction is rate-limiting
(b) The adsorption of isobutene is rate limiting
(c) The reaction follows Eley-Rideal kinetics
$W + I \bullet S \rightarrow TBA \bullet S$
and that the surface reaction is limiting.